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15-4c The Analytics of Price Discrimination

Let's consider a bit more formally how price discrimination affects economic welfare. We begin by assuming that the monopolist can price discriminate perfectly. *Perfect price discrimination* describes a situation in which the monopolist knows exactly each customer's willingness to pay and can charge each customer a different price. In this case, the monopolist charges each customer exactly her willingness to pay, and the monopolist gets the entire surplus in every transaction.

Figure 9 illustrates producer and consumer surplus with and without price discrimination. To keep things simple, this figure is drawn assuming constant per unit costs—that is, marginal cost and average total cost are constant and equal. Without price discrimination, the firm charges a single price above marginal cost, as shown in panel (a). Because some potential customers who value the good at more than marginal cost do not buy it at this high price, the monopoly causes a deadweight loss. Yet when a firm can perfectly price discriminate, as shown in panel (b), each customer who values the good at more than marginal cost buys the good and is charged her willingness to pay. All mutually beneficial trades take place, no deadweight loss occurs, and the entire surplus derived from the market goes to the monopoly producer in the form of profit.

Figure 9

Welfare with and without Price Discrimination

Panel (a) shows a monopoly that charges the same price to all customers. Total surplus in this market equals the sum of profit (producer surplus) and consumer surplus. Panel (b) shows a monopoly that can perfectly price discriminate. Because consumer surplus equals zero, total surplus now equals the firm's profit. Comparing these two panels, you can see that perfect price discrimination raises profit, raises total surplus, and lowers consumer surplus.

In reality, of course, price discrimination is not perfect. Customers do not walk into stores with signs displaying their willingness to pay. Instead, firms price discriminate by dividing customers into groups: young versus old, weekday versus weekend shoppers, Americans versus Australians, and so on. Unlike those in our parable of Readalot Publishing, customers within each group differ in their willingness to pay for the product, making perfect price discrimination impossible.

How does this imperfect price discrimination affect welfare? The analysis of these pricing schemes is complicated, and it turns out that there is no general answer to this question. Compared with the single-price monopoly outcome, imperfect price discrimination can raise, lower, or leave unchanged the total surplus in a market. The only certain conclusion is that price discrimination raises the monopoly's profit; otherwise, the firm would choose to charge all customers the same price.

Chapter 15: Monopoly: 15-4c The Analytics of Price Discrimination
Book Title: WGU Economics
Printed By: Mike Benton (mbento6@wgu.edu)
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